

▼ **Figure 17.25 A summary of transcription and translation in a eukaryotic cell.** This diagram shows the path from one gene to one polypeptide. Each gene in the DNA can be transcribed repeatedly into many identical RNA molecules, and each mRNA can be translated repeatedly to yield many identical polypeptide molecules. (Remember that the final products of some genes are not polypeptides but RNA molecules that don't get translated,

including tRNAs and rRNAs.) In general, the steps of transcription and translation are similar in bacterial, archaeal, and eukaryotic cells. The major difference is the occurrence of RNA processing in the eukaryotic nucleus. Other significant differences are found in the initiation stages of both transcription and translation and in the termination of transcription. To visualize these processes in their cellular context, see Figure 6.32d-f.

